

Deep Reconnaissance, Scanning & Vulnerability Intelligence Report

Project	: NovaShyld Task 2
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Assessment Type	: Vulnerability Assessment & Penetration Testing (VAPT) Lab Reconnaissance and Vulnerability finding
Attacker Machine	: Kali Linux 2026.x
Target Machine	: Metasploitable 2
Virtualization Platform	: Oracle VirtualBox

1. Objective

Perform advanced reconnaissance, correlate scan data, and identify realistic attack paths based on vulnerabilities

Module 1: Intelligent Reconnaissance

- Host discovery was performed using network scanning tools to identify active systems in the network.
- Banner grabbing was conducted using tools like **netcat** and **curl** to extract service information.
- Identified running services along with their versions to understand system exposure and attack surface.
- Analyzed service behavior to determine possible misconfigurations and vulnerabilities.

Module 2: Multi-Layer Scanning

Performed multiple Nmap scan types to gather comprehensive network intelligence:

- TCP SYN Scan (-sS): Stealth scan to detect open ports efficiently.
- UDP Scan (-sU): Identified UDP-based services and potential hidden services.

- Version Detection (-sV): Determined service versions running on open ports.
- OS Detection (-O): Estimated the operating system of the target machine.
- All scan outputs were saved to files for comparison and documentation.

Key Outcome:

- Differences between scan types helped reveal hidden services and improve accuracy of system profiling.

Module 3: Vulnerability Intelligence

- Performed vulnerability scanning using **OpenVAS tool**.
- Generated detailed vulnerability reports after scanning the target system.
- Focused on identifying:
 - Critical vulnerabilities
 - High severity vulnerabilities
- Studied how automated scanners detect vulnerabilities based on service fingerprints and known CVEs.

The screenshot displays the Greenbone Security Assistant (GSA) web interface in a Mozilla Firefox browser. The address bar shows the URL: `https://127.0.0.1:9392/report/fbe3f0df-bf87-4e13-8637-34fd...`. The interface has a green header with the Greenbone logo and a navigation menu with tabs: Dashboards, Scans, Assets, Resilience, SecInfo, Configuration, Administration, and Help.

The main content area shows a report for a scan named "metasploitable". The report header includes the scan name, a progress bar at 0%, and the scan time: "Thu, Jun 25, 2026 3:10 AM UTC". Below the header is a table with the following columns: Information, Results (0 of 0), Hosts (0 of 1), Ports (0 of 0), Applications (0 of 0), Operating Systems (0 of 0), CVEs (0 of 0), Closed CVEs (0 of 0), TLS Certificates (0 of 0), Error Messages (0 of 0), and User Tags (0).

Below the table, the "Task Name" is "metasploitable". The "Scan Time" is "Thu, Jun 25, 2026 3:10 AM UTC". The "Scan Status" is "0 %". The "Hosts scanned" is "1". The "Filter" is "apply_overrides=0 levels=hml min_qod=70". The "Timezone" is "Coordinated Universal Time (UTC)".

1 Result Overview

Host	High	Medium	Low	Log	False Positive
10.0.2.14	24	34	5	0	0
Total	1	24	34	5	0

Vendor security updates are not trusted.

Overrides are off. Even when a result has an override, this report uses the actual threat of the result.

Information on overrides is included in the report.

Notes are included in the report.

This report might not show details of all issues that were found.

Issues with the threat level "Log" are not shown.

Issues with the threat level "Debug" are not shown.

Issues with the threat level "False Positive" are not shown.

Only results with a minimum QoD of 70 are shown.

This report contains all 63 results selected by the filtering described above. Before filtering there were 531 results.

1.1 Host Authentications

Host	Protocol	Result	Port/ User
10.0.2.14	SMB	Success	Protocol SMB, Port 445, User

2 Results per Host

2.1 10.0.2.14

Host scan start Thu Jun 25 03:38:10 2026 UTC

Host scan end Thu Jun 25 04:38:08 2026 UTC

Service (Port)	Threat Level
6200/tcp	High
6697/tcp	High
8009/tcp	High
8787/tcp	High
21/tcp	High
513/tcp	High
5900/tcp	High
5432/tcp	High
22/tcp	High

...(continues) ...

...(continued) ...

Service (Port)	Threat Level
512/tcp	High
2121/tcp	High
3306/tcp	High
general/tcp	High
80/tcp	High
514/tcp	High
1524/tcp	High
3632/tcp	High
23/tcp	Medium
21/tcp	Medium
5900/tcp	Medium
5432/tcp	Medium
25/tcp	Medium
22/tcp	Medium
2121/tcp	Medium
445/tcp	Medium
80/tcp	Medium
5432/tcp	Low
25/tcp	Low
22/tcp	Low
general/tcp	Low

2. Executive Summary

- A Vulnerability Assessment was performed on the Metasploitable 2 virtual machine (10.0.2.14) using OpenVAS.
- The assessment identified multiple **Critical** and **High** severity vulnerabilities across network services and web applications.

- Critical vulnerabilities were detected on:
 - Port 1524 – Ingreslock Backdoor
 - Port 513 – rlogin
 - Port 3632 – DistCC
 - Port 8009 – Apache Tomcat AJP
 - Port 8787 – Distributed Ruby (DRb)
- High-severity vulnerabilities included:
 - Port 3306 – MySQL Weak Authentication
 - Port 5432 – PostgreSQL Weak Authentication
 - Port 5900 – VNC Weak Authentication
- Manual validation confirmed:
 - Direct root shell access through Port 1524.
 - Successful login via rlogin followed by privilege escalation to root using sudo.
- The identified vulnerabilities could allow attackers to gain unauthorized access, execute arbitrary commands, compromise sensitive information, and obtain complete control of the system.
- The overall security posture of the target system is **Critical**, requiring immediate remediation through patch management, removal of weak/default credentials, service hardening, network access restrictions, and continuous vulnerability management.

Attack Chain (Example Scenario)

A realistic exploitation path observed in the lab environment:

1. **Nmap scan** reveals open SMB (445) and Telnet (23)
2. SMB enumeration identifies vulnerable Samba version
3. Exploit executed → remote shell obtained
4. System access confirmed as low-privilege user
5. Privilege escalation achieved via local misconfiguration

6. Full root access obtained

Final Impact Assessment

If exploited in a real-world environment, these vulnerabilities could lead to:

- Full remote system compromise
- Unauthorized access to sensitive files
- Credential theft
- Network lateral movement
- Complete loss of confidentiality, integrity, and availability

3. Conclusion

The Metasploitable 2 system was found to have multiple critical and high-severity vulnerabilities due to outdated services, weak authentication, and exposed network ports. These issues allow potential unauthorized access, privilege escalation, and full system compromise. Overall, the system is assessed as **Critical risk**, requiring immediate hardening, patching, and removal of insecure services